

## Learning Objectives

This chapter contains the following topics:

- review of the structure of a tree, its leaf and non-leaf nodes, and the arrangement of arcs between these nodes
- decision trees and their utility in deciding the class of a given pattern
- associating each non-leaf node with an attribute being examined and each leaf node with a class or with rejection
- a breadth-first procedure to build a decision tree from a given training set
- building different decision trees from the professor-student training set of the first chapter
- showing how different decision trees built from the same training set can classify a given recall pattern differently
- developing the mathematical model for the ratio of information gain an attribute provides when examined at a node of a decision tree
- building a decision tree in which the attribute selected to be examined at a node maximizes the ratio of information gain
- obtaining production rules from a given decision tree

# 2

## Decision Trees: Basics

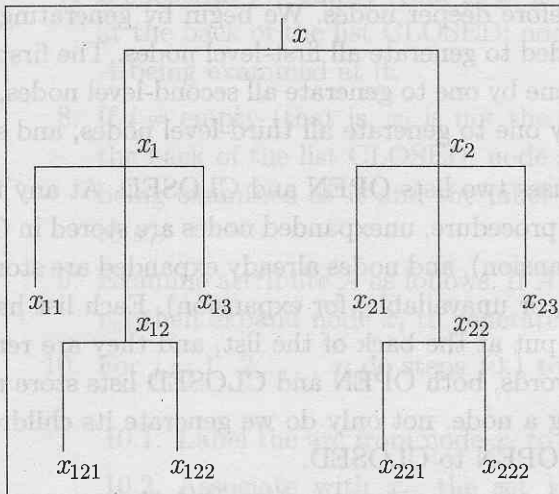
## 2.1 Trees

This section briefly discusses a data structure called a tree and the terminology used in trees, mainly to refresh the reader's memory and to remove any ambiguity about the terminology. It is advisable to periodically refer to Fig. 2.1, which illustrates a tree.

A *tree* is a set of one or more *nodes*. A node  $x_i$ , subscript  $i$  being either empty or a sequence of one or more non-negative integers, is joined to another node  $x_{ij}$  by an *arc* directed from  $x_i$  to  $x_{ij}$  (between two nodes there can be one arc at the most, and no arc is directed from a node to the node itself). Node  $x_{ij}$  is said to be a *child* of  $x_i$ , and  $x_i$  is the *parent* of  $x_{ij}$ . When subscript  $i$  is empty, an alternative way is not to write it; thus  $x_i$  may be written as  $x$ , and  $x_{ij}$  as  $x_j$ .

If a node  $x_i$  has  $n \geq 2$  children  $x_{i1}, x_{i2}, \dots, x_{in}$ , then the children are said to be one another's *siblings*. *Expanding* a node means generating its children. A node with no children is a *leaf* node; otherwise, it is a *non-leaf* node.

A node  $x_i$  is an *ancestor* of another node  $x_j$  if, and only if, either  $x_i$  is a parent of  $x_j$ , or  $x_i$  is a parent of an ancestor of  $x_j$ . No node is an ancestor



**FIGURE 2.1**

An illustration of a tree. The root node is  $x$ . Its two children are  $x_1$  and  $x_2$ , node  $x_1$  being the sibling of  $x_2$  and vice versa. The root node is the ancestor of every other node in the tree. The leaf nodes are  $x_{11}$ ,  $x_{13}$ ,  $x_{21}$ ,  $x_{23}$ ,  $x_{121}$ ,  $x_{122}$ ,  $x_{221}$ , and  $x_{222}$ . The other nodes  $x$ ,  $x_1$ ,  $x_2$ ,  $x_{12}$ , and  $x_{22}$  are non-leaf nodes. The level of root  $x$  is 0, and the level of nodes  $x_1$  and  $x_2$  is 1. The depth of the tree is 3.

of itself. A node  $x_j$  is a *descendant* of a node  $x_i$  if, and only if,  $x_i$  is an ancestor of  $x_j$ . The *root* node is the ancestor of all other nodes in the tree. In other words, all other nodes are descendants of the root. The sequence of arcs from a node to its descendants is a *path*.

The *depth* (or *level*) of a node is zero for a root node, and one more than the depth of its parent for a non-root node. If the depth of a node  $x_i$  is less than the depth of a node  $x_j$ , then  $x_i$  is *shallower* than  $x_j$ , and  $x_j$  is *deeper* than  $x_i$ . The *depth of a tree* is equal to the depth of its deepest node.

## 2.2 Building a Decision Tree

A *decision tree* is a kind of tree which, in the training phase is built from a training set. Then in the testing phase, the tree is used to classify patterns. We can build more than one decision tree from a given training set. The procedure that follows builds an arbitrary decision tree. Later, Section 2.5 develops a criterion for selecting the tree to be built.

This procedure builds the tree in a *breadth-first* manner: shallower nodes are generated before deeper nodes. We begin by generating the root node. It is then expanded to generate all first-level nodes. The first-level nodes are then expanded one by one to generate all second-level nodes, which are then expanded one by one to generate all third-level nodes, and so on.

The procedure uses two lists OPEN and CLOSED. At any time during the execution of the procedure, unexpanded nodes are stored in OPEN (they are still open to expansion), and nodes already expanded are stored in CLOSED (they are closed, or unavailable, for expansion). Each list has a front and a back: nodes are put at the back of the list, and they are removed from the front. In other words, both OPEN and CLOSED lists store nodes as queues do. In expanding a node, not only do we generate its children, but we also transfer it from OPEN to CLOSED.

Let  $V$  denote a training set. As mentioned in Section 1.1, let us assume that the training patterns have at least one attribute, each attribute has at least two possible values, the number of classes is  $m \geq 1$ , and each pattern belongs to exactly one class. The breadth-first procedure to build a decision tree from a given training set is as follows.

1. If  $m = 1$  (that is, there is only one class), then create a single node  $x$ , label  $x$  with the name of the single class, and terminate the procedure. The decision tree consists of the single node  $x$ , with its class label.
2. Initialize lists OPEN and CLOSED to empty.
3. Initialize subscript  $i$  to empty. Create a root node  $x_i$ , and associate the training set  $V_i$  with it.
4. Put node  $x_i$  in OPEN.
5. If OPEN is empty, return from the procedure. The nodes in CLOSED constitute the decision tree built. The subscript and the label of each node in CLOSED, together with the labelled arc from the node's parent, delineate the decision tree (step 10.1 explains how the arcs are labelled).
6. Remove the frontmost node  $x_i$  from OPEN. Create a *candidate set of attributes* for  $x_i$ , where the set contains all those attributes that have not been examined at any node on the path from the root to  $x_i$ . Select an attribute  $A$  from the candidate set for examining at node  $x_i$  (the criterion developed in Section 2.5 can be used to select attributes; for now, it may be selected arbitrarily).
7. If subscript  $i = \text{empty}$  (that is,  $x_i$  is the root), then put the following at the back of the list CLOSED: node  $x_i$  together with the attribute  $A$  being examined at it.
8. If  $i \neq \text{empty}$  (that is,  $x_i$  is not the root), then put the following at the back of the list CLOSED: node  $x_i$  together with the attribute  $A$  being examined at it and the label of the arc from the parent-of- $x_i$  to  $x_i$ .
9. Examine attribute  $A$  as follows. If  $A$  has  $n$  possible values  $v_1, v_2, \dots, v_n$ , then expand node  $x_i$  to generate its  $n$  children  $x_{i1}, x_{i2}, \dots, x_{in}$ .
10. For  $j = 1, 2, \dots, n$  do steps 10.1 to 10.8.
  - 10.1. Label the arc from node  $x_i$  to node  $x_{ij}$  with attribute value  $v_j$ .
  - 10.2. Associate with  $x_{ij}$  the set  $V_{ij} \subseteq V_i$ , such that the value of attribute  $A$  for the patterns in  $V_{ij}$  is  $v_j$ .
  - 10.3. If all the patterns in set  $V_{ij}$  belong to one class, then label node  $x_{ij}$  with the name of that class. Go to step 10.7.
  - 10.4. If  $V_{ij}$  is empty, then label node  $x_{ij}$  with '?' to indicate rejection, that is, failure to classify a given pattern. Go to step 10.7. (The question mark symbol, '?', has been used to indicate rejection,

which means that the question of pattern's class still exists. You can replace it by a symbol of your choice.)

- 10.5. If the patterns in  $V_{ij}$  belong to more than one class and all the attributes have been examined on the path from the root to  $x_i$ , then label  $x_{ij}$  with '?' to indicate rejection. (One could say that ideally we should have more attributes available to put the patterns of  $V_{ij}$  into separate classes, but more attributes may not be available. An alternative to rejection is to label node  $x_{ij}$  with the probability of the occurrence of different classes in  $V_{ij}$ ; more on this is given in Chapter 3.) Go to step 10.7.
- 10.6. Put the following at the back of the list OPEN: the node  $x_{ij}$  (it is a non-leaf node), and the label of the arc from node  $x_i$  to  $x_{ij}$ . Go to step 10.8.
- 10.7. Put the following at the back of the list CLOSED: node  $x_{ij}$ , a mark to indicate that  $x_{ij}$  is a leaf node, the label of  $x_{ij}$ , and the label of the arc from  $x_i$  to  $x_{ij}$ .
- 10.8. Continue.

# 11. Go to step 5.

On the path from the root to a leaf node, no attribute is examined more than once. It is, however, not necessary that every attribute be examined before we arrive at a leaf node, as can happen in steps 10.3 and 10.4 of this procedure. The depth of a leaf node will therefore be less than or equal to the number of attributes. Since, by definition, the depth of a tree is equal to the depth of its deepest node, the depth of the tree, too, will be less than or equal to the number of attributes.

As an example, let us build a decision tree using the professor-student training set given in Fig. 1.1. Of the eight patterns in the training set, patterns T2, T4, and T7 belong to the class P (for professor), and patterns T1, T3, T5, T6, and T8 belong to the class S (for student). We associate training set  $V$  with node  $x$ , which will become the root of the decision tree. To understand this better, it is advisable to write down the contents of the training set  $V$  next to node  $x$  given in Fig. 2.1.

The candidate set of attributes at node  $x$  consists of HABIT, EATS, and FOOTWEAR. Let us—for the time being, arbitrarily—select HABIT to be

examined at node  $x$  (the criterion for selecting an attribute is developed in Section 2.5). Attribute HABIT has two possible values: gabby and quiet. We expand  $x$  to generate its two children  $x_1$  and  $x_2$ . We label the arc from  $x$  to  $x_1$  as gabby, and the arc from  $x$  to  $x_2$  as quiet.

We have labelled the arc from  $x$  to  $x_1$  as gabby, so we associate set  $V_1 = \{T1, T2, T3, T5, T7\}$  with node  $x_1$  (because from the set  $V$ , the patterns in  $V_1$  have HABIT equal to gabby). Similarly, since we have labelled the arc from  $x$  to  $x_2$  as quiet, we associate set  $V_2 = \{T4, T6, T8\}$  with node  $x_2$  (because from the set  $V$ , the patterns in  $V_2$  have HABIT equal to quiet).

Since HABIT has already been examined at the parent of node  $x_1$ , the candidate set of attributes at  $x_1$  consists of EATS and FOOTWEAR. Let us, arbitrarily, select EATS to be examined at  $x_1$ . Attribute EATS has three possible values: baked, fried, and roasted. We expand  $x_1$  to generate its three children  $x_{11}$ ,  $x_{12}$ , and  $x_{13}$ , and label the three arcs from  $x_1$  to its children as baked, fried, and roasted, respectively. We associate set  $V_{11} = \{T1, T3\}$  with node  $x_{11}$  (because from the set  $V_1$ , the patterns in  $V_{11}$  have EATS equal to baked). Since all the patterns in  $V_{11}$  belong to the class student, node  $x_{11}$  becomes a leaf node (see step 10.3 of the above procedure to build decision trees), and we label it with S (for student).

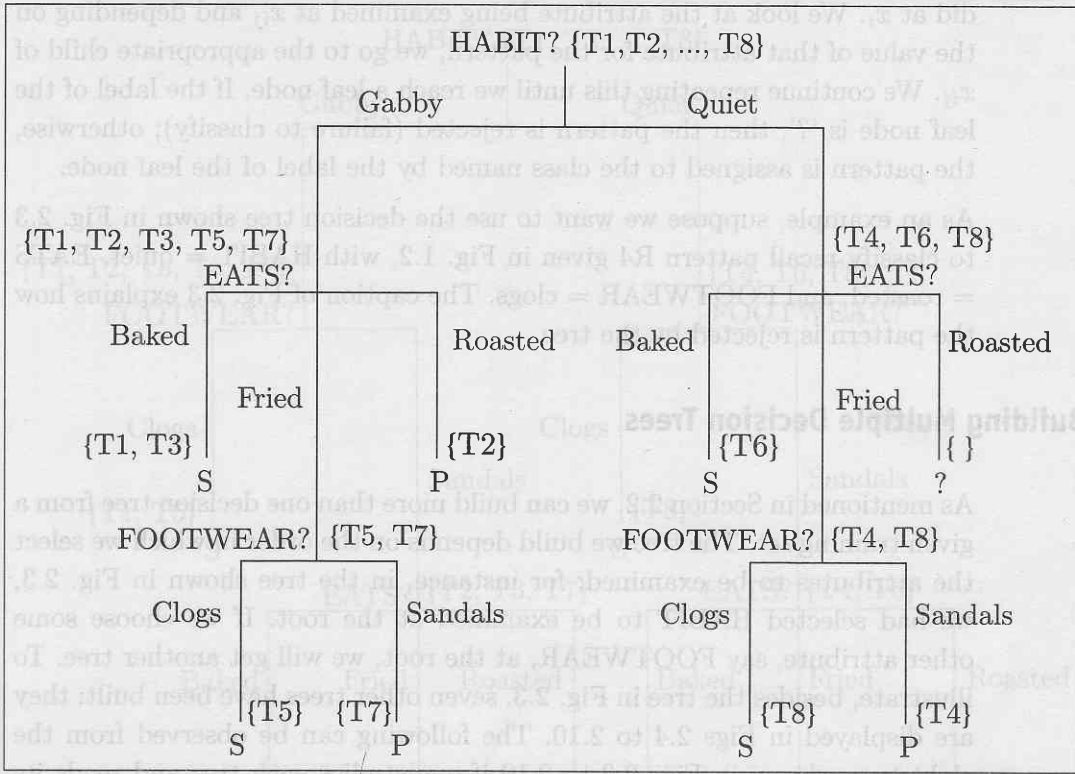
The reader can complete the decision tree for this example following the procedure mentioned above. Figure 2.2 contains a brief explanation for generating each node of the tree and can be referred to if there is any difficulty in completing the decision tree.

The decision tree built should finally look like the tree given in Fig. 2.3. The following minor changes have been made to this tree to make it easier to understand.

- The names of the nodes ( $x, x_1, x_2, \dots$ ) and the names of the sets ( $V, V_1, V_2, \dots$ ) associated with the nodes have not been shown to avoid cluttering the tree.
- At a non-leaf node, a question mark has been put after the name of the attribute being examined to indicate that the question of the class of the pattern has not yet been answered, and that we are looking at the possible values of that attribute.

| Node       | Set associated with the node     | Attribute examined                                  | Arc from node to |
|------------|----------------------------------|-----------------------------------------------------|------------------|
| $x$ (root) | {T1, T2, T3, T4, T5, T6, T7, T8} | HABIT = gabby                                       | $x_1$            |
|            |                                  | = quiet                                             | $x_2$            |
| $x_1$      | {T1, T2, T3, T5, T7}             | EATS = baked                                        | $x_{11}$         |
|            |                                  | = fried                                             | $x_{12}$         |
|            |                                  | = roasted                                           | $x_{13}$         |
| $x_2$      | {T4, T6, T8}                     | EATS = baked                                        | $x_{21}$         |
|            |                                  | = fried                                             | $x_{22}$         |
|            |                                  | = roasted                                           | $x_{23}$         |
| $x_{11}$   | {T1, T3}                         | None, since node becomes a leaf node with label S   | No other node    |
| $x_{12}$   | {T5, T7}                         | FOOTWEAR = clogs                                    | $x_{121}$        |
|            |                                  | = sandals                                           | $x_{122}$        |
| $x_{13}$   | {T2}                             | None, since node becomes a leaf node with label P   | No other node    |
| $x_{21}$   | {T6}                             | None, since node becomes a leaf node with label S   | No other node    |
| $x_{22}$   | {T4, T8}                         | FOOTWEAR = clogs                                    | $x_{221}$        |
|            |                                  | = sandals                                           | $x_{222}$        |
| $x_{23}$   | { }                              | None, since node becomes a leaf node with label '?' | No other node    |
| $x_{121}$  | {T5}                             | None, since node becomes a leaf node with label S   | No other node    |
| $x_{122}$  | {T7}                             | None, since node becomes a leaf node with label P   | No other node    |
| $x_{221}$  | {T8}                             | None, since node becomes a leaf node with label S   | No other node    |
| $x_{222}$  | {T4}                             | None, since node becomes a leaf node with label P   | No other node    |

**FIGURE 2.2** Generating nodes for a decision tree from the professor-student training set given in Fig. 1.1. The symbols '{ }' denote an empty set and '?' a rejection. A leaf node is labelled S for the class student and P for the class professor. The decision tree is shown in Fig. 2.3.



**FIGURE 2.3** Decision tree built from the professor–student training set given in Fig. 1.1. Figure 2.2 shows how each node was obtained. At a leaf node, P denotes the class professor and S denotes the class student. To classify the recall pattern R4 of Fig. 1.2 (HABIT = quiet, EATS = roasted, and FOOTWEAR = clogs), we start from the root and examine the value of HABIT for the pattern; since the value is quiet, we arrive at the right child of the root. At the child, we examine the value of EATS; since it is roasted, we arrive at the leaf node labelled '?'. The pattern is rejected even without examining its attribute FOOTWEAR.

### 2.3 Classifying by Using Decision Trees

Suppose we are given a pattern (be it recall or training), which means we are given the value of its various attributes and we want to classify the pattern by using an available decision tree.

We start from the root of the tree. At any non-leaf node  $x_i$ , we look at the attribute being examined. Suppose the possible values of that attribute are  $v_1, v_2, \dots, v_n$ , and the value of the attribute for the pattern is  $v_j$ . Then we go to  $x_{ij}$ , the  $j$ th child of  $x_i$ . If  $x_{ij}$  is a non-leaf node, we repeat what we

did at  $x_i$ . We look at the attribute being examined at  $x_{ij}$  and depending on the value of that attribute for the pattern, we go to the appropriate child of  $x_{ij}$ . We continue repeating this until we reach a leaf node. If the label of the leaf node is '?', then the pattern is rejected (failure to classify); otherwise, the pattern is assigned to the class named by the label of the leaf node.

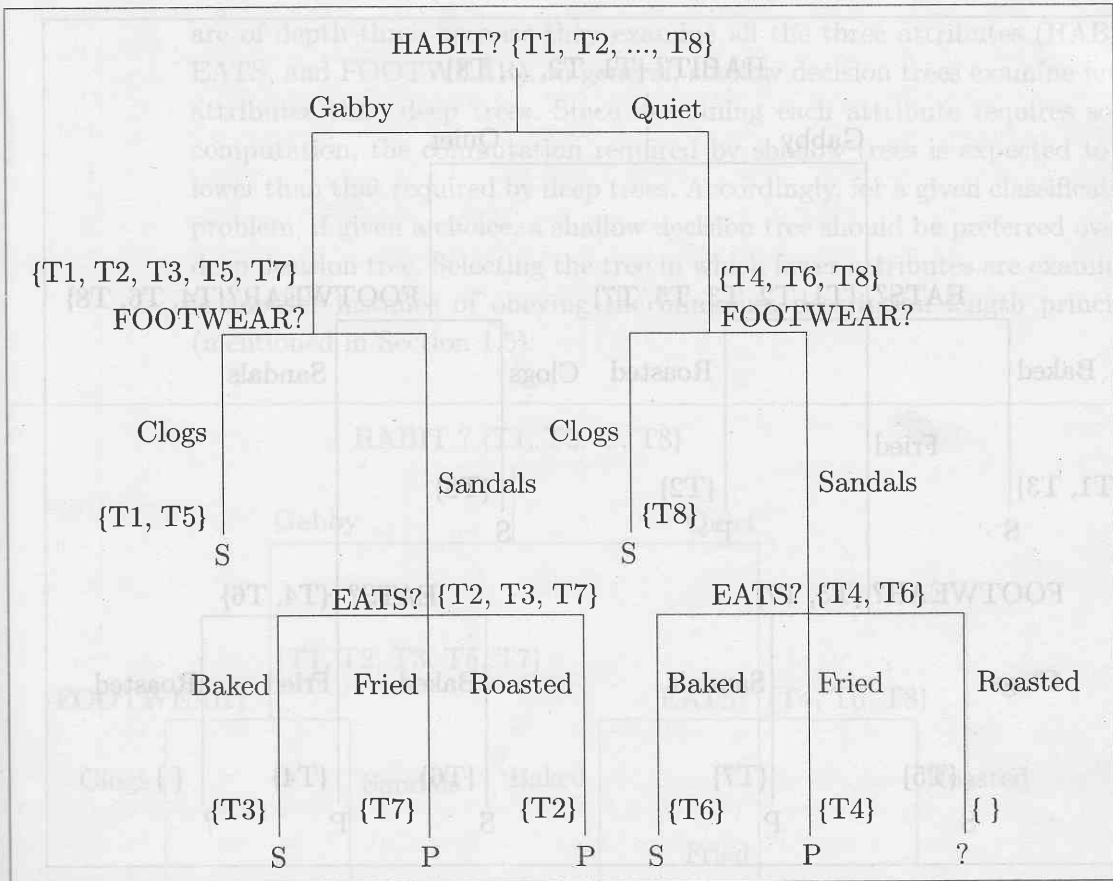
As an example, suppose we want to use the decision tree shown in Fig. 2.3 to classify recall pattern R4 given in Fig. 1.2, with HABIT = quiet, EATS = roasted, and FOOTWEAR = clogs. The caption of Fig. 2.3 explains how the pattern is rejected by the tree.

## 2.4 Building Multiple Decision Trees

As mentioned in Section 2.2, we can build more than one decision tree from a given training set. The tree we build depends on the order in which we select the attributes to be examined; for instance, in the tree shown in Fig. 2.3, we had selected HABIT to be examined at the root. If we choose some other attribute, say FOOTWEAR, at the root, we will get another tree. To illustrate, besides the tree in Fig. 2.3, seven other trees have been built: they are displayed in Figs 2.4 to 2.10. The following can be observed from the eight trees shown in Figs 2.3 to 2.10 if one studies each tree and reads its caption.

- It is not necessary that the same attribute be examined at sibling nodes: in Fig. 2.5, at level 1, one node examines EATS, while the other node examines FOOTWEAR. Similarly, the sibling nodes in Fig. 2.6 examine different attributes.
- The three decision trees in Figs 2.7 to 2.9 do not reject any pattern, but the remaining trees do.
- The two decision trees shown in Figs 2.7 and 2.9 are of depth two and the remaining trees are of depth three.

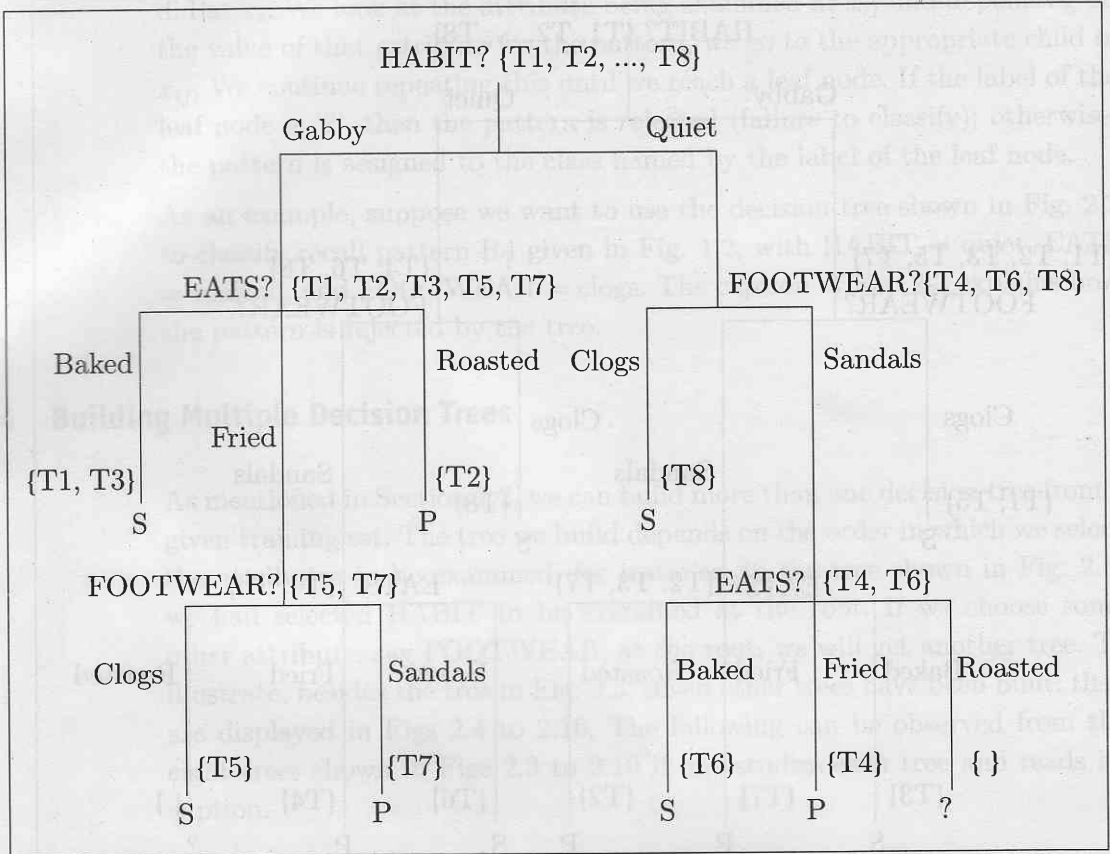
From the captions of the decision trees given in Figs 2.3 to 2.10, we can see that, as an example, when we classify recall pattern R4 given in Fig. 1.2 (HABIT = quiet, EATS = roasted, and FOOTWEAR = clogs) using the eight different trees, we do not obtain an identical classification—Figs 2.3 and 2.6 reject it; Figs 2.4, 2.5, 2.9, and 2.10 classify it in the class student; and Figs 2.7 and 2.8 classify it in the class professor. In general, a given pattern can be classified differently by different decision trees. The classifications of the four recall patterns given in Fig. 1.2 according to the



**FIGURE 2.4** Second decision tree obtained from the professor–student training set given in Fig. 1.1. The recall pattern R4 given in Fig. 1.2 (HABIT = quiet, EATS = roasted, and FOOTWEAR = clogs) is classified by the above tree as student. This is different from the result given by the decision tree in Fig. 2.3, where the pattern is rejected.

eight trees are given in Fig. 2.11. It is useful to verify these results by working on them. Observe that recall pattern R1 has been classified as student by all the eight trees, but the classifications of patterns R2 to R4 vary.

It will not be surprising if different people are asked to classify intuitively the recall patterns given in Fig. 1.2, and they come out with different results. One can assume that they have intuitively built different decision trees.

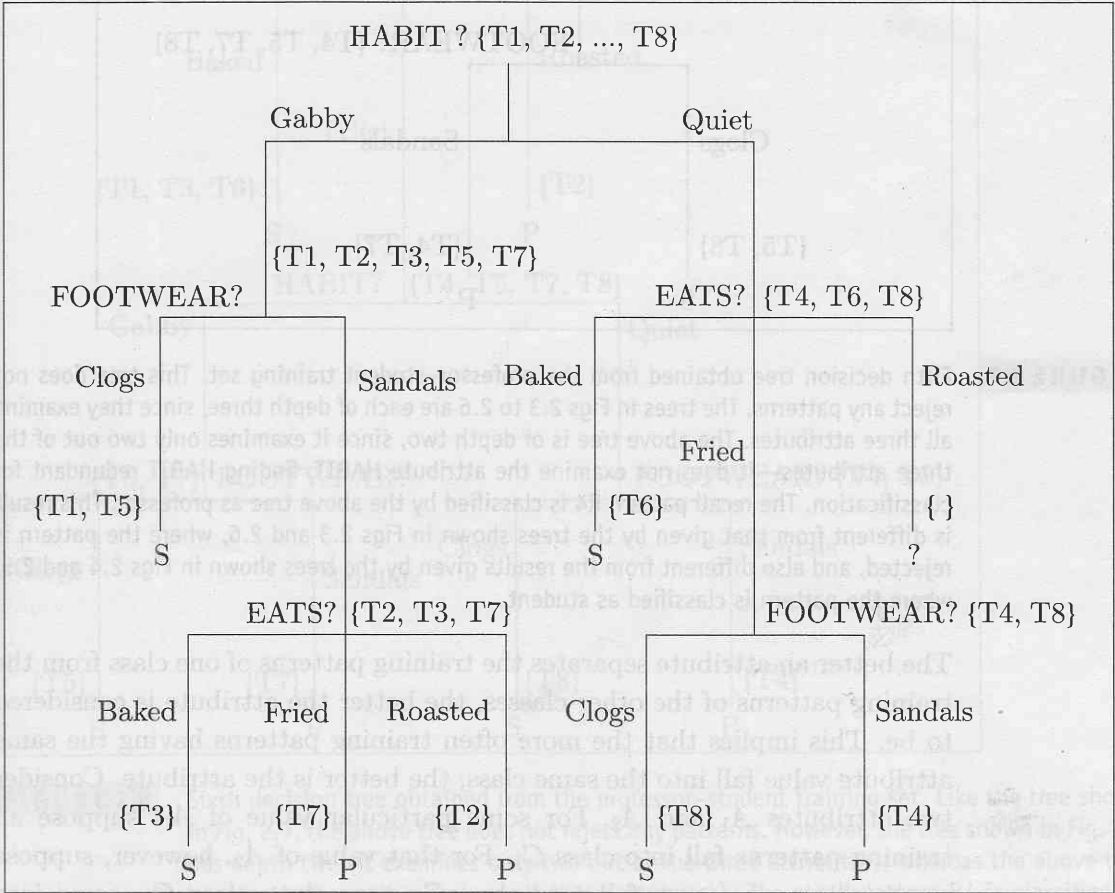


**FIGURE 2.5** Third decision tree obtained from the professor–student training set. It is not necessary that the same attribute be examined at sibling nodes—in this figure, at depth 1, we examine EATS at one node, and FOOTWEAR at its sibling. The recall pattern R4 is classified by the above tree as student. This result is the same as given by the decision tree shown in Fig. 2.4, but different from the result given by the decision tree shown in Fig. 2.3, where the pattern is rejected.

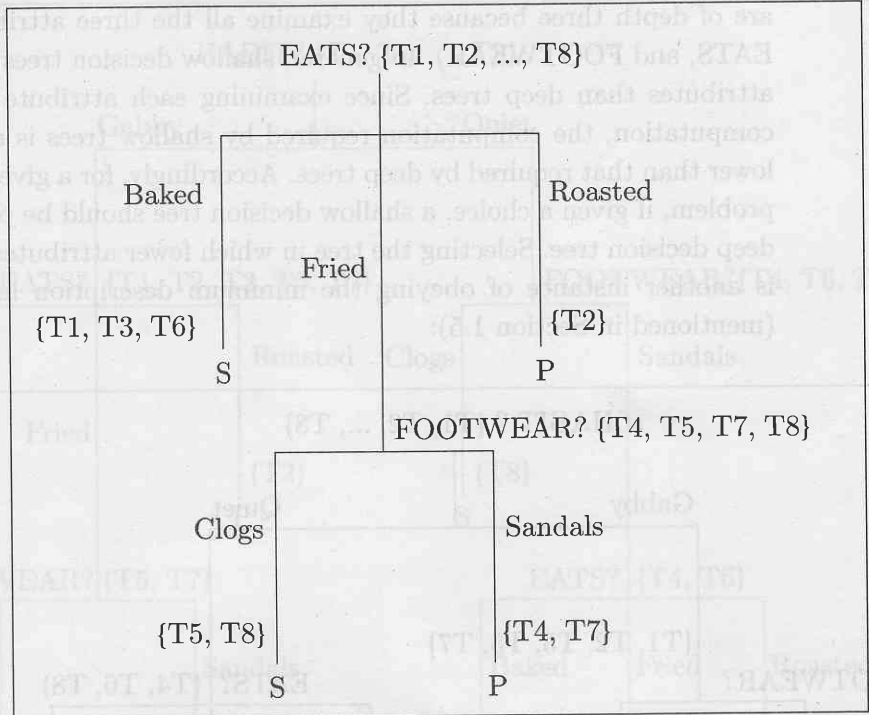
## 2.5 Selecting the Decision Tree to be Built

While reading the professor–student training set example in Chapter 1, it might not have become immediately evident that attribute HABIT is redundant and that it need not be examined to classify a pattern. The decision trees shown in Figs 2.7 and 2.9, however, confirm it. These decision trees are of depth two because they examine only two attributes (EATS and FOOTWEAR). The other six decision trees (Figs 2.3 to 2.6, 2.8, and 2.10)

are of depth three because they examine all the three attributes (HABIT, EATS, and FOOTWEAR). In general, shallow decision trees examine fewer attributes than deep trees. Since examining each attribute requires some computation, the computation required by shallow trees is expected to be lower than that required by deep trees. Accordingly, for a given classification problem, if given a choice, a shallow decision tree should be preferred over a deep decision tree. Selecting the tree in which fewer attributes are examined is another instance of obeying the minimum description length principle (mentioned in Section 1.5):



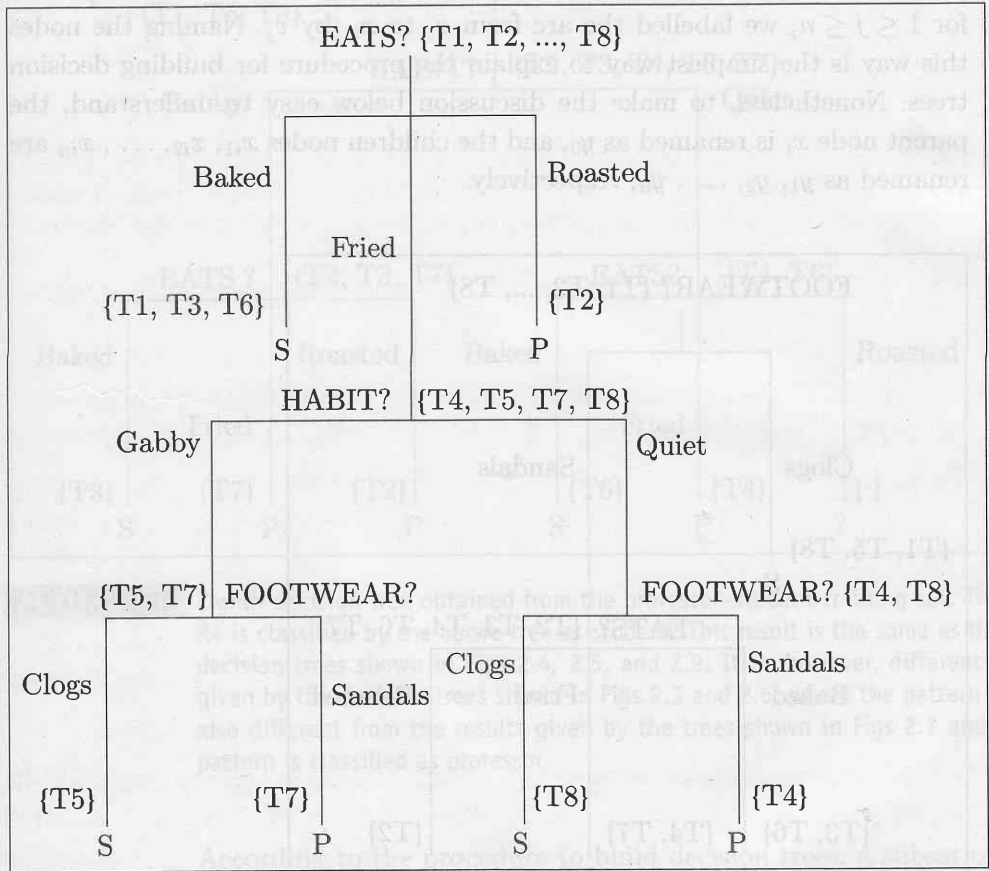
**FIGURE 2.6** Fourth decision tree obtained from the professor–student training set. The recall pattern R4 is rejected by the above tree. This result is the same as that given by the decision tree shown in Fig. 2.3, but different from that given by the decision trees shown in Figs 2.4 and 2.5, where the pattern is classified as student.

**FIGURE 2.7**

Fifth decision tree obtained from the professor–student training set. This tree does not reject any patterns. The trees in Figs 2.3 to 2.6 are each of depth three, since they examine all three attributes. The above tree is of depth two, since it examines only two out of the three attributes—it does not examine the attribute HABIT, finding HABIT redundant for classification. The recall pattern R4 is classified by the above tree as professor. This result is different from that given by the trees shown in Figs 2.3 and 2.6, where the pattern is rejected, and also different from the results given by the trees shown in Figs 2.4 and 2.5, where the pattern is classified as student.

The better an attribute separates the training patterns of one class from the training patterns of the other classes, the better the attribute is considered to be. This implies that the more often training patterns having the same attribute value fall into the same class, the better is the attribute. Consider two attributes  $A_1$  and  $A_2$ . For some particular value of  $A_1$ , suppose  $n$  training patterns fall into class  $C_1$ . For that value of  $A_2$ , however, suppose  $m$  training patterns fall into class  $C_1$ , some into class  $C_2$ , some into class  $C_3$ , and so on. Then for this attribute value,  $A_1$  separates the training patterns better than  $A_2$ . If  $A_1$  does so for other values as well, then attribute  $A_1$  is better than attribute  $A_2$ . In general, the more unevenly an attribute distributes training patterns over different classes, the better the attribute.

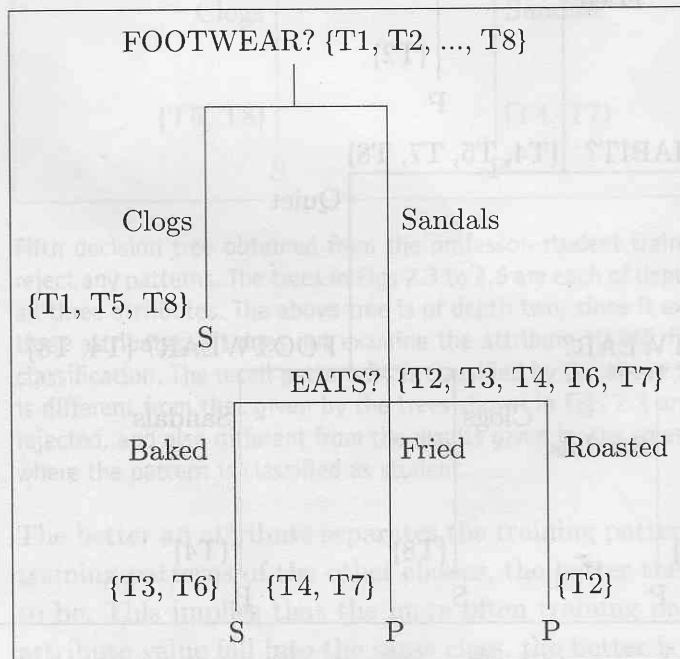
Whenever we examine an attribute at some node of a decision tree, we gain some information about the class of a given pattern. The information gained is a function of the unevenness with which the attribute distributes the training patterns over the different classes. Better attributes provide a higher ratio of information gain compared with other attributes. From the candidate set of attributes for a given non-leaf node in a decision tree, we can choose to examine the attribute that maximizes this ratio. A decision



**FIGURE 2.8** Sixth decision tree obtained from the professor-student training set. Like the tree shown in Fig. 2.7, the above tree does not reject any patterns. However, the tree shown in Fig. 2.7 has depth two (it examines only two out of the three attributes), whereas the above tree has depth three (it examines all three attributes). The recall pattern R4 is classified by the above tree as professor. This result is the same as that given by the decision tree shown in Fig. 2.7, but different from that given by the tree shown in Figs 2.3 and 2.6, where the pattern is rejected, and also different from the results given by the trees of Figs 2.4 and 2.5, where the pattern is classified as student.

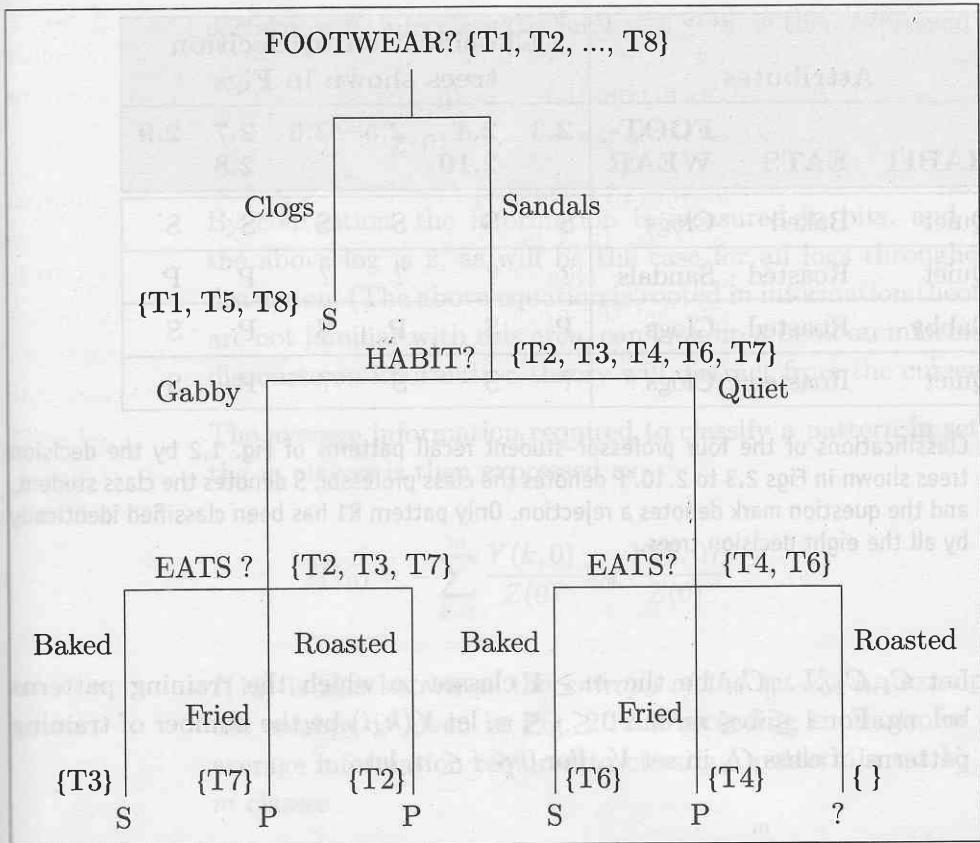
tree built by maximizing this ratio for each of the tree's non-leaf nodes is typically shallower than a tree built otherwise. The description below develops a mathematical equation to evaluate this ratio for a given attribute  $A$  being examined at some node.

In step 9 of the procedure to build decision trees (Section 2.2), we examined attribute  $A$  at a node  $x_i$ . If  $A$  has  $n$  possible values  $v_1, v_2, \dots, v_n$ , we expand  $x_i$  to generate its  $n$  children  $x_{i1}, x_{i2}, \dots, x_{in}$ . Then, in step 10.1, for  $1 \leq j \leq n$ , we labelled the arc from  $x_i$  to  $x_{ij}$  by  $v_j$ . Naming the nodes this way is the simplest way to explain the procedure for building decision trees. Nonetheless, to make the discussion below easy to understand, the parent node  $x_i$  is renamed as  $y_0$ , and the children nodes  $x_{i1}, x_{i2}, \dots, x_{in}$  are renamed as  $y_1, y_2, \dots, y_n$ , respectively.



**FIGURE 2.9**

Seventh decision tree obtained from the professor-student training set. Like the tree shown in Fig. 2.7, the above tree is also of depth two (it examines only two out of the three attributes) and it does not reject any patterns. The recall pattern R4 is classified by the above tree as student. This result is the same as given by the decision trees shown in Figs 2.4 and 2.5, but different from that given by the decision trees shown in Figs 2.3 and 2.6, where the pattern is rejected, and also different from that given by the trees shown in Figs 2.7 and 2.8, where the pattern is classified as professor.



**FIGURE 2.10** Eighth decision tree obtained from the professor–student training set. The recall pattern R4 is classified by the above tree as student. This result is the same as that given by the decision trees shown in Figs 2.4, 2.5, and 2.9. It is, however, different from the result given by the decision trees shown in Figs 2.3 and 2.6, where the pattern is rejected, and also different from the results given by the trees shown in Figs 2.7 and 2.8, where the pattern is classified as professor.

According to the procedure to build decision trees, a subset of the training set is associated with each node: for  $0 \leq i \leq n$ , a set  $V_i$  is associated with node  $y_i$ . At node  $y_0$ , we examine attribute  $A$  on the patterns of set  $V_0$ . Then, for  $1 \leq j \leq n$ , we have  $V_j \subseteq V_0$ , such that the value of attribute  $A$  for the patterns in  $V_j$  is  $v_j$ , where  $v_j$  is the label of the arc from node  $y_0$  to node  $y_j$ . Set  $V_0$  is thus the union of the sets  $V_1$  to  $V_n$ . We can alternatively say that attribute  $A$  *splits* set  $V_0$  into sets  $V_1$  to  $V_n$ .

| Name of recall pattern | Attributes |         |           | Classification by decision trees shown in Figs |             |     |     |            |     |
|------------------------|------------|---------|-----------|------------------------------------------------|-------------|-----|-----|------------|-----|
|                        | HABIT      | EATS    | FOOT-WEAR | 2.3                                            | 2.4<br>2.10 | 2.5 | 2.6 | 2.7<br>2.8 | 2.9 |
| R1                     | Quiet      | Baked   | Clogs     | S                                              | S           | S   | S   | S          | S   |
| R2                     | Quiet      | Roasted | Sandals   | ?                                              | ?           | ?   | ?   | P          | P   |
| R3                     | Gabby      | Roasted | Clogs     | P                                              | S           | P   | S   | P          | S   |
| R4                     | Quiet      | Roasted | Clogs     | ?                                              | S           | S   | ?   | P          | S   |

**FIGURE 2.11** Classifications of the four professor–student recall patterns of Fig. 1.2 by the decision trees shown in Figs 2.3 to 2.10. P denotes the class professor, S denotes the class student, and the question mark denotes a rejection. Only pattern R1 has been classified identically by all the eight decision trees.

Let  $C_1, C_2, \dots, C_m$  be the  $m \geq 1$  classes to which the training patterns belong. For  $1 \leq k \leq m$  and  $0 \leq i \leq n$ , let  $Y(k, i)$  be the number of training patterns of class  $C_k$  in set  $V_i$ . For  $0 \leq i \leq n$ , let

$$Z(i) = \sum_{k=1}^m Y(k, i)$$

which means that  $Z(i)$  is equal to the number of patterns in the set  $V_i$ , that is,  $Z(i)$  is equal to the cardinality of  $V_i$ .

We can now develop the equation for the ratio of information gain provided by attribute  $A$  when it is examined at node  $y_0$ . For  $1 \leq k \leq m$ , the probability that a pattern in  $V_0$  belongs to class  $C_k$  is

$$\frac{Y(k, 0)}{Z(0)}$$

The lower the above probability, the more the uncertainty in classifying a pattern in set  $V_0$  into class  $C_k$ . The more the uncertainty, the more the information required to classify. Thus, the lower the above probability, the more the need for such information. The information required to classify a

pattern in  $V_0$  into class  $C_k$ , for  $1 \leq k \leq m$ , is then expressed as

$$-\log \frac{Y(k, 0)}{Z(0)}$$

By convention, the information is measured in bits, and so the base of the above log is 2, as will be the case for all logs throughout the current discussion. (The above equation is rooted in information theory; readers who are not familiar with this area, can look up a book on information theory; A discourse on information theory will distract from the current discussion.)

The average information required to classify a pattern in set  $V_0$  into one of the  $m$  classes is then expressed as

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} \quad (2.1)$$

$I(V_0)$  is also known as the *entropy of the set*  $V_0$ . An example to calculate the entropy is given in Fig. 2.12. Proceeding similarly for  $1 \leq j \leq n$ , the average information required to classify a pattern in set  $V_j$  into one of the  $m$  classes

$$I(V_j) = - \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)}$$

The average information required to classify a pattern belonging to set  $V_0$  after it has been split by attribute  $A$  into sets  $V_1$  to  $V_n$  is

$$I_A(V_0) = \sum_{j=1}^n \frac{Z(j)}{Z(0)} I(V_j)$$

Combining the last two equations, we get

$$I_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} \quad (2.2)$$

To evaluate the entropy  $I(V_0)$  of the professor–student training set of Fig. 1.1. The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Three patterns belong to  $C_1$  and five to  $C_2$ . Therefore,

The training set  $V_0 = \{T1, T2, \dots, T8\}$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

From Eqn (2.1) in Section 2.5, the average information required to classify a pattern in  $V_0$  into one of the  $m$  classes is

$$\begin{aligned}
 I(V_0) &= - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} \\
 &= - \sum_{k=1}^2 \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} \\
 &= -\frac{3}{8} \log \frac{3}{8} - \frac{5}{8} \log \frac{5}{8} \\
 &= 0.375 \times 1.415 + 0.625 \times 0.6780 \\
 &= 0.5306 + 0.4238 \\
 &= 0.9544
 \end{aligned}$$

**FIGURE 2.12** Evaluating entropy  $I(V_0)$ , where  $V_0$  is the professor–student training set of Fig. 1.1.

$I_A(V_0)$  is also known as the *entropy of the attribute A* for set  $V_0$ . The *gain in information* caused by attribute  $A$  splitting set  $V_0$  into sets  $V_1$  to  $V_n$  is then defined as

$$g_A(V_0) = I(V_0) - I_A(V_0) \quad (2.3)$$

We want to maximize this gain in information. For a given non-leaf node, we calculate the gain in information for each attribute in the node's candidate set of attributes. The attribute that has the highest gain can then be selected

for examination at the node. This criterion of selection can be used in step 6 of the procedure to build decision trees, described in Section 2.2.

There are, however, occasions when maximizing this gain can be misleading. The professor-student training patterns given in Fig. 1.1 have been arbitrarily given the names T1 to T8. Suppose we were to consider NAME to be an attribute. At the root, NAME will split set  $V_0$  into eight sets  $V_1$  to  $V_8$ , where each  $V_j$ , for  $1 \leq j \leq 8$ , will contain one pattern. The patterns in sets  $V_2, V_4$ , and  $V_7$  will belong to class  $C_1$  (professor), and the patterns in the remaining sets will belong to class  $C_2$  (student). The entropy of the attribute NAME will be zero, and consequently it will maximize the gain in information. Nonetheless, the attribute NAME will be useless in classifying recall patterns—since each pattern has a unique NAME value, no recall pattern will have a NAME value that is identical to the NAME value of a training pattern. A decision tree built by examining NAME at the root will reject all recall patterns. For an attribute to be useful, patterns should share attribute values. By maximizing information gain, we bias our selection in favour of the attribute that splits set  $V_0$  into more non-empty subsets and against the attribute that splits  $V_0$  into fewer non-empty subsets.

To rectify this bias, we need to also consider the attribute's *split information* (how the attribute splits set  $V_0$ ). The probability that there are  $Z(j)$  patterns of set  $V_j$  among the  $Z(0)$  patterns of set  $V_0$  is

$$\frac{Z(j)}{Z(0)}$$

The information needed to extract set  $V_j$  from  $V_0$  is

$$-\log \frac{Z(j)}{Z(0)}$$

So, the average information needed by attribute  $A$  to split set  $V_0$  into sets  $V_1$  to  $V_n$  is

$$S_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} \quad (2.4)$$

The *ratio of information gain* of attribute  $A$  for set  $V_0$  is then defined as

$$G_A(V_0) = \frac{g_A(V_0)}{S_A(V_0)}$$

Then, from Eqn (2.3), we get the ratio of information gain

$$G_A(V_0) = \frac{I(V_0) - I_A(V_0)}{S_A(V_0)} \quad (2.5)$$

We want to maximize this ratio of information gain. For a given non-leaf node, we calculate this ratio for each attribute in the node's candidate set of attributes. The attribute that has the highest ratio is then selected for examination at the node. This criterion of selection can be used in step 6 of the procedure to build decision trees, described in Section 2.2.

When we build a decision tree, as the nodes become deeper and deeper, the training set gets split further and further. Thus, the deeper the non-leaf node, the fewer the training patterns associated with it, and less is the theoretical justification in using the ratio of information gain to decide on the attribute to be examined at that node. In practice, however, the ratio is used at deep nodes as well. As an example, we will use the ratio to build the decision tree for the professor-student training set given in Fig. 1.1.

1. The candidate set of attributes that can be examined at the root consists of HABIT, EATS, and FOOTWEAR. From Figs 2.13 to 2.15, we see that, at the root,  $G_{\text{HABIT}}(V_0) = 0.0047$ ,  $G_{\text{EATS}}(V_0) = 0.3233$ , and  $G_{\text{FOOTWEAR}}(V_0) = 0.3640$ . Since  $G_{\text{FOOTWEAR}}(V_0)$  is the largest of the three ratios of information gain, we select FOOTWEAR to be examined at the root. Accordingly, the decision tree we will build will look either like the tree in Fig. 2.9, or like the tree in Fig. 2.10.
2. Examining FOOTWEAR at the root, we notice, from Figs 2.9 and 2.10, that the set associated with the root's left child is {T1, T5, T8}. Every pattern in the set belongs to the class student; hence the left child becomes a leaf node labelled S for student. The set associated with the root's right child is then {T2, T3, T4, T6, T7}, out of which two patterns T3 and T6 belong to the class student, and the remaining three patterns belong to the class professor.
3. The candidate set of attributes that can be examined at the root's right child consists of HABIT and EATS, attribute FOOTWEAR having been already examined at the root. From Figs 2.16 and 2.17 we see that, at the right child of the root,  $G_{\text{EATS}}(V_0) = 0.6377$  and  $G_{\text{HABIT}}(V_0) = 0.0208$ . Since  $G_{\text{EATS}}(V_0)$  is the larger of the two ratios

of information gain, we select EATS to be examined at the right child of the root. The nodes generated by expanding the root's right child are all leaf nodes, and we obtain the decision tree shown in Fig. 2.9.

As the above example shows, using the criterion of maximizing ratio of information gain to build a decision tree, may require considerable computation. This criterion can still be recommended because such computation is required to be carried out only once to build the decision tree. Such a tree, however, is expected to save a lot more computation when the tree is later repeatedly used to classify recall patterns. The learning program mentioned in Section 1.1 is the one that will build the appropriate decision tree, for a given training set.

## 2.6 Obtaining Prules From Decision Trees

The learning program is often extended to obtain prules from the decision tree built. Patterns are then classified using these prules. Each path from the root to a leaf node provides a prule. The consequent of the prule contains the label of the leaf node. The antecedent of the prule is the conjunction of the attribute values required to reach that leaf node. In Section 2.5, the decision tree shown in Fig. 2.9 was built by maximizing the ratio of information gain. The following four prules constitute the knowledge base obtained from this tree.

If FOOTWEAR = clogs,  
then the pattern belongs to the class student.

If FOOTWEAR = sandals,  
and EATS = baked,  
then the pattern belongs to the class student.

If FOOTWEAR = sandals,  
and EATS = fried,  
then the pattern belongs to the class professor.

If FOOTWEAR = sandals,  
and EATS = roasted,  
then the pattern belongs to the class professor.

To evaluate  $G_{\text{HABIT}}(V_0)$  at the root of a decision tree for the professor–student training set of Fig. 1.1, the following procedure is used. The root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since HABIT has two possible values (gabby and quiet), root  $y_0$  has two children  $y_1$  and  $y_2$ . Therefore,  $n = 2$ . To understand the following material easily, one can look at any one of the decision trees shown in Figs 2.3 to 2.6, since each of them examines HABIT at the root.

$V_0 = \{T1, T2, \dots, T8\}$  at node  $y_0$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

$V_1 = \{T1, T2, T3, T5, T7\}$  at node  $y_1$ , where HABIT = gabby

$Y(1, 1) = 2$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2, 1) = 3$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 5$  (number of patterns in  $V_1$ )

$V_2 = \{T4, T6, T8\}$  at node  $y_2$ , where HABIT = quiet

$Y(1, 2) = 1$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2, 2) = 2$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 3$  (number of patterns in  $V_2$ )

$I(V_0) = 0.9544$  (from Fig. 2.12)

$$I_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.9499 \quad [\text{from Eqn (2.2)}]$$

$$S_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 0.9544 \quad [\text{from Eqn (2.4)}]$$

$$G_{\text{HABIT}}(V_0) = \frac{I(V_0) - I_{\text{HABIT}}(V_0)}{S_{\text{HABIT}}(V_0)} = 0.0047 \quad [\text{from Eqn (2.5)}]$$

**FIGURE 2.13** Evaluating  $G_{\text{HABIT}}(V_0)$ , the ratio of information gain of HABIT, at the root of a decision tree being built from the professor–student training set shown in Fig. 1.1.

To evaluate  $G_{\text{EATS}}(V_0)$  at the root of a decision tree for the professor–student training set, the following procedure is used. The root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since EATS has three possible values (baked, fried, and roasted), root  $y_0$  has three children  $y_1$ ,  $y_2$ , and  $y_3$ . Therefore,  $n = 3$ . To understand the following material easily, one can look at either of the decision trees shown in Figs 2.7 or 2.8, since each of them examines EATS at the root.

$V_0 = \{T1, T2, \dots, T8\}$  at node  $y_0$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

$V_1 = \{T1, T3, T6\}$  at node  $y_1$ , where EATS = baked

$Y(1, 1) = 0$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2, 1) = 3$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 3$  (number of patterns in  $V_1$ )

$V_2 = \{T4, T5, T7, T8\}$  at node  $y_2$ , where EATS = fried

$Y(1, 2) = 2$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2, 2) = 2$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 4$  (number of patterns in  $V_2$ )

$V_3 = \{T2\}$  at node  $y_3$ , where EATS = roasted

$Y(1, 3) = 1$  (number of patterns in  $V_3$  of class  $C_1$ )

$Y(2, 3) = 0$  (number of patterns in  $V_3$  of class  $C_2$ )

$Z(3) = 1$  (number of patterns in  $V_3$ )

$I(V_0) = 0.9544$  (from Fig. 2.12)

$$I_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.5 \text{ [from Eqn (2.2)]}$$

$$S_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.4055 \text{ [from Eqn (2.4)]}$$

$$G_{\text{EATS}}(V_0) = \frac{I(V_0) - I_{\text{EATS}}(V_0)}{S_{\text{EATS}}(V_0)} = 0.3233 \text{ [from Eqn (2.5)]}$$

**FIGURE 2.14** Evaluating  $G_{\text{EATS}}(V_0)$ , the ratio of information gain of EATS, at the root of a decision tree built from the professor–student training set.

To evaluate  $G_{\text{FOOTWEAR}}(V_0)$  at the root of a decision tree for the professor-student training set, the following procedure is used. The root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since FOOTWEAR has two possible values (clogs and sandals), root  $y_0$  has two children  $y_1$  and  $y_2$ . Therefore,  $n = 2$ . To understand the following material easily, one can look at either of the decision trees shown in Fig. 2.9 or 2.10, since each of them examines FOOTWEAR at the root.

$V_0 = \{T1, T2, \dots, T8\}$  at node  $y_0$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

$V_1 = \{T1, T5, T8\}$  at node  $y_1$ , where FOOTWEAR = clogs

$Y(1, 1) = 0$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2, 1) = 3$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 3$  (number of patterns in  $V_1$ )

$V_2 = \{T2, T3, T4, T6, T7\}$  at node  $y_2$ , where FOOTWEAR = sandals

$Y(1, 2) = 3$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2, 2) = 2$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 5$  (number of patterns in  $V_2$ )

$I(V_0) = 0.9544$  (from Fig. 2.12)

$$I_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.6066 \text{ [from Eqn (2.2)]}$$

$$S_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 0.9544 \text{ [from Eqn (2.4)]}$$

$$G_{\text{FOOTWEAR}}(V_0) = \frac{I(V_0) - I_{\text{FOOTWEAR}}(V_0)}{S_{\text{FOOTWEAR}}(V_0)} = 0.3640 \text{ [from Eqn (2.5)]}$$

**FIGURE 2.15** Evaluating  $G_{\text{FOOTWEAR}}(V_0)$ , the ratio of information gain of FOOTWEAR, at the root of a decision tree being built from the professor-student training set.

To evaluate  $G_{\text{EATS}}(V_0)$  at the right child of the root of a decision tree for the professor–student training set, the following procedure is used. FOOTWEAR has already been examined at the root (see Fig. 2.9). The right child of the root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since EATS has three possible values (baked, fried, and roasted), node  $y_0$  has three children  $y_1$ ,  $y_2$ , and  $y_3$ . Therefore,  $n = 3$ .

$V_0 = \{\text{T2, T3, T4, T6, T7}\}$  at root  $y_0$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 2$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 5$  (number of patterns in  $V_0$ )

$V_1 = \{\text{T3, T6}\}$  at node  $y_1$ , where EATS = baked

$Y(1, 1) = 0$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2, 1) = 2$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 2$  (number of patterns in  $V_1$ )

$V_2 = \{\text{T4, T7}\}$  at node  $y_2$ , where EATS = fried

$Y(1, 2) = 2$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2, 2) = 0$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 2$  (number of patterns in  $V_2$ )

$V_3 = \{\text{T2}\}$  at node  $y_3$ , where EATS = roasted

$Y(1, 3) = 1$  (number of patterns in  $V_3$  of class  $C_1$ )

$Y(2, 3) = 0$  (number of patterns in  $V_3$  of class  $C_2$ )

$Z(3) = 1$  (number of patterns in  $V_3$ )

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} = 0.9710 \quad [\text{from Eqn (2.1)}]$$

$$I_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0 \quad [\text{from Eqn (2.2)}]$$

$$S_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.5218 \quad [\text{from Eqn (2.4)}]$$

$$G_{\text{EATS}}(V_0) = \frac{I(V_0) - I_{\text{EATS}}(V_0)}{S_{\text{EATS}}(V_0)} = 0.6377 \quad [\text{from Eqn (2.5)}]$$

**FIGURE 2.16** Evaluating  $G_{\text{EATS}}(V_0)$ , the ratio of information gain of EATS, at the right child of the root of a decision tree being built from the professor–student training set, FOOTWEAR having already been examined at the root, as in Fig. 2.9.

To evaluate  $G_{\text{HABIT}}(V_0)$  at the right child of the root of a decision tree for the professor-student training set, the following procedure is used. FOOTWEAR has already been examined at the root (see Fig. 2.10). The right child of the root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since HABIT has two possible values (gabby and quiet), node  $y_0$  has two children  $y_1$  and  $y_2$ . Therefore,  $n = 2$ .

$V_0 = \{T2, T3, T4, T6, T7\}$  at node  $y_0$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 2$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 5$  (number of patterns in  $V_0$ )

$V_1 = \{T2, T3, T7\}$  at node  $y_1$ , where HABIT = gabby

$Y(1, 1) = 2$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2, 1) = 1$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 3$  (number of patterns in  $V_1$ )

$V_2 = \{T4, T6\}$  at node  $y_2$ , where HABIT = quiet

$Y(1, 2) = 1$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2, 2) = 1$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 2$  (number of patterns in  $V_2$ )

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} = 0.9710 \text{ [from Eqn (2.1)]}$$

$$I_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.9507 \text{ [from Eqn (2.2)]}$$

$$S_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 0.9710 \text{ [from Eqn (2.4)]}$$

$$G_{\text{HABIT}}(V_0) = \frac{I(V_0) - I_{\text{HABIT}}(V_0)}{S_{\text{HABIT}}(V_0)} = 0.0208 \text{ [from Eqn (2.5)]}$$

**FIGURE 2.17** Evaluating  $G_{\text{HABIT}}(V_0)$ , the ratio of information gain of HABIT, at the right child of root of a decision tree being built from the professor-student training set, FOOTWEAR having already been examined at the root, as in Fig. 2.10.

A default prule (see Section 1.1) is not needed because the decision tree shown in Fig. 2.9, from which the above prules are obtained, does not reject any patterns. The recall pattern R4 given in Fig. 1.2 (HABIT = quiet, EATS = roasted, and FOOTWEAR = clogs) is classified as student, by firing the first prule. This classification is the same as that given by the decision tree shown in Fig. 2.9.

In general, a decision tree and the prules obtained from that tree should provide the same results for a given pattern. Using prules is usually preferred because the prules present the knowledge used in classification in a form that is easy to read and understand. For the learning program, building a decision tree from a given training set thus becomes a step in obtaining prules for that set.

You may want to compare the above prules obtained from Fig. 2.9 with those obtained by the SpecToGen procedure (Fig. 1.3) and those obtained by the GenToSpec procedure (Fig. 1.5).

## Summary

A decision tree built from a training set can later be used to classify training and recall patterns. Since we can build multiple decision trees from a given training set, we select a decision tree, in which, at every non-leaf node, the attribute examined maximizes the ratio of information gain. The prules obtained from the decision tree built can be used for classification. Overall, a learning program should build the appropriate decision tree and then, if needed, obtain prules from it.

## Exercises

1. In a programming language of your choice, implement the procedure to build decision trees described in this chapter. The user should be able to specify the criterion that will be used to select the attribute to be examined at a non-leaf node: (a) arbitrarily, (b) by maximizing  $g_A(V_0)$ , the gain in information defined in Eqn (2.3), or (c) by maximizing  $G_A(V_0)$ , the ratio of information gain defined in Eqn (2.5).
2. In the procedure to build decision trees (Section 2.2), the set  $V_i$  associated with a node  $x_i$  is split into subsets  $V_{i1}, V_{i2}, \dots, V_{in}$ , which are then associated with nodes  $x_{i1}, x_{i2}, \dots, x_{in}$ . Then each

of the subsets  $V_{i1}, V_{i2}, \dots, V_{in}$  are similarly split. Can this be done recursively? If so, in a programming language of your choice implement a procedure that uses recursion to build decision trees. The user should be able to specify the criterion for selecting the attribute to be examined at a non-leaf node as in the first exercise.

3. Write a program in a language of your choice that, given a decision tree built by the first or second exercise above, will be able to classify a given pattern (see Section 2.3).
4. Obtain prules from each of the decision trees shown in Figs 2.3 to 2.10 and Fig. 2.10; the prules from Fig. 2.9 have already been given in Section 2.6. Classify patterns of the recall set shown in Fig. 1.2 using each of these seven knowledge bases (the sets of prules). Display your results in a table. Check whether your results match those given in Fig. 2.11, which were obtained by classifying the recall patterns directly from the corresponding decision trees. Is any set of prules obtained from the decision trees identical to either of the sets of prules given in Figs 1.3 or 1.5, which were obtained, respectively, by the SpecToGen and GenToSpec procedures of Sections 1.4 and 1.5?
5. Discuss the advantages and disadvantages of classifying patterns directly from decision trees versus first obtaining prules from the trees and then classifying the patterns using the prules. Remember that in domains of practical interest, decision trees may have depths of 10 or more; and a knowledge base may have many hundred prules, if not more.
6. The example solved in Section 2.5 maximizes  $G_A(V_0)$ , the ratio of information gain shown in Eqn (2.5), to build the decision tree from the professor-student training set given in Fig. 1.1. Would you build the same decision tree if you, instead, maximized  $g_A(V_0)$ , the gain in information shown in Eqn (2.3)? If not, which tree would you build? In maximizing  $g_A(V_0)$ , do you need to evaluate  $I(V_0)$ , or can that be avoided? If so, why and how can it be avoided?
7. Let the set of attributes consist of HABIT, EATS, FOOTWEAR, and NAME for the professor-student training set given in Fig. 1.1. Build a decision tree by each of these two methods.
  - (a) Maximize  $G_A(V_0)$ , the ratio of information gain
  - (b) Maximize  $g_A(V_0)$ , the gain in information

Compare the results obtained by the two methods. Which method would you recommend? Why?

- 8. To maximize the ratio of information gain (Section 2.5), all equations require log to the base 2. Many calculators and programming languages do not have functions built in to evaluate logs to the base 2. However, they do have logs to the base 10 and to the base e. We can find logs to the base 2 by either of the following equations:

$$\log_2 y = 3.3219 \times \log_{10} y$$

$$\log_2 y = 1.4427 \times \log_e y$$

If we were to use either logs to the base 10 or logs to the base e to maximize the ratio of information gain in selecting the attribute to be examined, will we get different results? Discuss.

- 9. As you may be aware, a deck of playing cards has four suits: clubs, spades, diamonds, and hearts. The colour of clubs and spades is black, and that of diamonds and hearts is red. Each suit has 13 cards: cards one to ten are called *numbered* cards, and cards jack, queen, and king are called *court* or *face* cards. Build a decision tree from the training set given in Fig. 2.18, and then identify the classes of the following cards or reject them: king of hearts, three of diamonds, jack of clubs, queen of diamonds, six of spades, nine of hearts, king of spades, and five of clubs.
- 10. Measure twenty computer science departments that carry out both teaching and research, on the following attributes (the students are

| NAME of<br>card   | Attributes |          | Class          |
|-------------------|------------|----------|----------------|
|                   | COLOUR     | TYPE     |                |
| Jack of hearts    | Red        | Court    | C <sub>1</sub> |
| Nine of spades    | Black      | Numbered | C <sub>2</sub> |
| King of clubs     | Black      | Court    | C <sub>2</sub> |
| Queen of diamonds | Red        | Court    | C <sub>1</sub> |
| Six of clubs      | Black      | Numbered | C <sub>2</sub> |

FIGURE 2.18 Training set for playing cards.

postgraduate—Master's and doctoral—students, the faculty are those who teach these students and supervise their theses, and averages are over the last five years for the department).

- (a) Average number of students in the department
- (b) Average number of faculty in the department
- (c) Average age of the faculty in the department (this is considered because the older faculty may have more experience)
- (d) Average number of research papers published in journals every year by the department faculty and students
- (e) Average monetary value (rupees, dollars, francs, or whatever be the currency unit in your country) of research and industrial grants received by the department every year
- (f) Average number of doctoral theses accepted every year

Allocate ten departments into class  $C_1$  and the remaining ten into class  $C_2$ , the  $C_1$  departments being intuitively considered to be better than the  $C_2$  departments. Arbitrarily, select seven departments from  $C_1$  and seven from  $C_2$  to form a training set. Put the remaining three departments each from  $C_1$  and  $C_2$  into a recall set. From a decision tree, develop prules to classify patterns of the training set into classes  $C_1$  and  $C_2$ . Test the prules on the patterns of the recall set and report how many of the recall patterns were classified correctly. You may collect data for this exercise from what you know about computer science departments in some universities, or you may thoughtfully create data on your own. Solve this exercise doing all the calculations manually, or use the programs you wrote for the first three exercises.

| Class | COLOUR | SUIT              |
|-------|--------|-------------------|
| 1     | Red    | Jack of hearts    |
| 2     | Black  | Nine of spades    |
| 3     | Black  | Count             |
| 4     | Red    | Queen of diamonds |
| 5     | Black  | Six of clubs      |